

Linear Regression Models Recitation Week 4

Question 1 (Theorem 2.8)

Let $\hat{\beta}_1$ and $\hat{\beta}_0$ be the least squares estimators of β_1 and β_0 . Then

$$\text{Cov}(\hat{\beta}_1, \hat{\beta}_0) = -\bar{x} \text{Var}[\hat{\beta}_1].$$

Solution. Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n,$$

with independent errors satisfying

$$E(\varepsilon_i) = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2.$$

Direct Proof via Linear Combinations of the Y_i 's

Using bilinearity of covariance,

$$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \text{Cov}(\bar{Y} - \bar{x}\hat{\beta}_1, \hat{\beta}_1) = \text{Cov}(\bar{Y}, \hat{\beta}_1) - \bar{x} \text{Var}(\hat{\beta}_1).$$

Write both quantities as linear combinations of the Y_i 's:

$$\bar{Y} = \sum_{i=1}^n a_i Y_i, \quad a_i = \frac{1}{n},$$

$$\hat{\beta}_1 = \sum_{j=1}^n c_j Y_j, \quad c_j = \frac{x_j - \bar{x}}{S_{xx}}.$$

By bilinearity of covariance,

$$\text{Cov}(\bar{Y}, \hat{\beta}_1) = \text{Cov}\left(\sum_i a_i Y_i, \sum_j c_j Y_j\right) = \sum_{i=1}^n \sum_{j=1}^n a_i c_j \text{Cov}(Y_i, Y_j).$$

Since the Y_i 's are independent,

$$\text{Cov}(Y_i, Y_j) = \begin{cases} \sigma^2, & i = j, \\ 0, & i \neq j, \end{cases}$$

so only the diagonal terms $i = j$ survive:

$$\text{Cov}(\bar{Y}, \hat{\beta}_1) = \sum_{i=1}^n a_i c_i \sigma^2 = \frac{\sigma^2}{n} \sum_{i=1}^n c_i.$$

Now

$$\sum_{i=1}^n c_i = \frac{1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) = \frac{1}{S_{xx}} (n\bar{x} - n\bar{x}) = 0.$$

Therefore

$$\text{Cov}(\bar{Y}, \hat{\beta}_1) = 0.$$

Consequently,

$$\boxed{\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = 0 - \bar{x} \text{Var}(\hat{\beta}_1) = -\bar{x} \text{Var}[\hat{\beta}_1].}$$

□

Proof via the Matrix Formulation (Optional)

The design matrix is

$$\mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \quad \hat{\boldsymbol{\beta}} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix}.$$

The least squares estimator satisfies

$$\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}.$$

Compute

$$\mathbf{X}^T \mathbf{X} = \begin{pmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{pmatrix} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum_{i=1}^n x_i^2 \end{pmatrix}.$$

The determinant is

$$\det(\mathbf{X}^T \mathbf{X}) = n \sum_{i=1}^n x_i^2 - n^2 \bar{x}^2 = n \left(\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right) = nS_{xx},$$

where

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2.$$

Therefore,

$$(\mathbf{X}^T \mathbf{X})^{-1} = \frac{1}{nS_{xx}} \begin{pmatrix} \sum_{i=1}^n x_i^2 & -n\bar{x} \\ -n\bar{x} & n \end{pmatrix} = \begin{pmatrix} \frac{\sum_{i=1}^n x_i^2}{nS_{xx}} & -\frac{\bar{x}}{S_{xx}} \\ -\frac{\bar{x}}{S_{xx}} & \frac{1}{S_{xx}} \end{pmatrix}.$$

Reading off the entries of $\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$:

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \cdot \frac{\sum_{i=1}^n x_i^2}{nS_{xx}}, \quad \text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}},$$

and the off-diagonal entry is

$$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \sigma^2 \left(-\frac{\bar{x}}{S_{xx}} \right) = -\bar{x} \frac{\sigma^2}{S_{xx}}.$$

Since $\text{Var}(\hat{\beta}_1) = \sigma^2/S_{xx}$, we obtain

$$\boxed{\text{Cov}(\hat{\beta}_1, \hat{\beta}_0) = -\bar{x} \text{Var}[\hat{\beta}_1].}$$

□

Question 2 (Homework 1 Problem 5)

Recall that the i -th fitted value $\hat{Y}_i$ can be expressed as a linear combination of the response values, i.e.,

$$\hat{Y}_i = \sum_{j=1}^n h_{ij} Y_j,$$

where

$$h_{ij} = \frac{1}{n} + \frac{(x_i - \bar{x})(x_j - \bar{x})}{S_{xx}}, \quad S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2.$$

Prove the following properties of the hat-values h_{ij} .

5.i: $\sum_{j=1}^n h_{ij}^2 = h_{ii}$

Solution.

$$\begin{aligned} \sum_{j=1}^n h_{ij}^2 &= \sum_{j=1}^n \left(\frac{1}{n} + \frac{(x_i - \bar{x})(x_j - \bar{x})}{S_{xx}} \right)^2 \\ &= \sum_{j=1}^n \left(\frac{1}{n^2} + \frac{2(x_i - \bar{x})(x_j - \bar{x})}{nS_{xx}} + \frac{(x_i - \bar{x})^2(x_j - \bar{x})^2}{S_{xx}^2} \right) \\ &= \sum_{j=1}^n \frac{1}{n^2} + \frac{2(x_i - \bar{x})}{nS_{xx}} \sum_{j=1}^n (x_j - \bar{x}) + \frac{(x_i - \bar{x})^2}{S_{xx}^2} \sum_{j=1}^n (x_j - \bar{x})^2 \\ &= \frac{1}{n} + 0 + \frac{(x_i - \bar{x})^2}{S_{xx}^2} S_{xx} \\ &= \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}} \\ &= h_{ii}. \end{aligned}$$

□

5.ii: $\sum_{j=1}^n h_{ij}x_j = x_i$

Solution.

$$\begin{aligned}
 \sum_{j=1}^n h_{ij}x_j &= \sum_{j=1}^n \left(\frac{1}{n} + \frac{(x_i - \bar{x})(x_j - \bar{x})}{S_{xx}} \right) x_j \\
 &= \frac{1}{n} \sum_{j=1}^n x_j + \frac{(x_i - \bar{x})}{S_{xx}} \sum_{j=1}^n (x_j - \bar{x})x_j \\
 &= \bar{x} + \frac{(x_i - \bar{x})}{S_{xx}} S_{xx} \\
 &= \bar{x} + (x_i - \bar{x}) \\
 &= x_i.
 \end{aligned}$$

□

Question 3 (Homework 1 Problem 6)

Consider the regression through the origin model given by

$$Y_i = \beta x_i + \epsilon_i, \quad i = 1, 2, \dots, n, \quad \epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2). \quad (1)$$

Here

$$\hat{y} = \hat{\beta}x, \quad \text{where} \quad \hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}.$$

Complete the following tasks.

6.i: Show that $\hat{\beta} = \frac{\sum_{i=1}^n x_i Y_i}{\sum_{i=1}^n x_i^2}$ is an unbiased estimator of β .

Solution. Note that under model (1),

$$E[Y_i] = E[\beta x_i + \epsilon_i] = \beta x_i + E[\epsilon_i] = \beta x_i.$$

Then

$$E[\hat{\beta}] = E\left[\frac{\sum_{i=1}^n x_i Y_i}{\sum_{i=1}^n x_i^2}\right] = \frac{\sum_{i=1}^n x_i E[Y_i]}{\sum_{i=1}^n x_i^2} = \frac{\sum_{i=1}^n x_i \cdot \beta x_i}{\sum_{i=1}^n x_i^2} = \beta \frac{\sum_{i=1}^n x_i^2}{\sum_{i=1}^n x_i^2} = \beta.$$

Hence $\hat{\beta}$ is unbiased for β .

□

6.ii: Find the standard error of the estimator $\hat{\beta}$.

Solution. Notice that $\hat{\beta}$ can be written as a linear combination of the response values Y_i . Namely,

$$\hat{\beta} = \sum_{i=1}^n c_i Y_i, \quad \text{where} \quad c_i = \frac{x_i}{\sum_{i=1}^n x_i^2}.$$

Also note that

$$\text{Var}[Y_i] = \text{Var}[\beta x_i + \epsilon_i] = \text{Var}[\epsilon_i] = \sigma^2.$$

Then, by independence of the Y_i 's,

$$\text{Var}[\hat{\beta}] = \text{Var}\left[\sum_{i=1}^n c_i Y_i\right] = \sum_{i=1}^n c_i^2 \text{Var}[Y_i] = \sum_{i=1}^n \left(\frac{x_i}{\sum_{i=1}^n x_i^2}\right)^2 \sigma^2 = \sigma^2 \frac{\sum_{i=1}^n x_i^2}{\left(\sum_{i=1}^n x_i^2\right)^2} = \frac{\sigma^2}{\sum_{i=1}^n x_i^2}.$$

Thus, the standard error of $\hat{\beta}$ is

$$\sigma_{\hat{\beta}} = \sqrt{\frac{\sigma^2}{\sum_{i=1}^n x_i^2}}.$$

□

6.iii: Identify the probability distribution of the estimator $\hat{\beta}$.

Solution. Since $\hat{\beta}$ is a linear combination of normal random variables,

$$\hat{\beta} \sim N\left(\beta, \frac{\sigma^2}{\sum_{i=1}^n x_i^2}\right).$$

□